MILITARY SPECIFICATION

TEST SET, FINAL ASSEMBLY: T4014A (PACKAGE 1 of 2) AND AUXILIARIES (PACKAGE 2 OF 2)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test set entitled T4014A Final Assembly Test Set which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-45889	- Cable Assemblies, W2-H, W3-H, W4-H, W6-H, W7-H, W8-H, and W20-H.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8785831	- Unit Assembly.
8803976	- Test Set, F.A.T., T4014 (T229) Assembly.
8803977	- Container, Metal, Ammunition, for Test Set, F.A.T., T4014 (T229), with Auxiliaries, Package 2 of 2.
8804125	- Assembly, Cover Removed.
8815011	- Dielectric Strength Tester.
EL-8815532	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawing (dwgs) 8803976 and 8803977, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Continuity. The test set (package 1 of 2) shall exhibit continuity and shall operate in accordance with the applicable drawing when tested as specified in 4.4.1.

3.6 Dielectric strength. The assembly shall show no evidence of damage or dielectric failure as evidenced by arcing between electrically-isolated parts or an inability to attain or maintain an alternating current (ac) potential of 500 ± 50 volts root mean square (vrms) at 60 ± 5 Hertz(Hz) for a duration of not less than 5 seconds when tested as specified in 4.4.2.

3.7 Insulation resistance. The insulation resistance between electrically-isolated circuits and between ungrounded circuits shall be not less than 20 megohms when subjected to a direct current (dc) potential of 500 ± 50 volts when tested as specified in 4.4.3.

3.8 Operating environment testing.

3.8.1 High temperature. The test set shall comply with the requirements specified in 3.5, 3.6, 3.7 and the applicable drawings while operating at 165° ± 5°F. when tested as specified in 4.4.4.1.

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After the completion of the test the test set shall show no evidence of deterioration or damage, such as obliterated markings, embrittled, distorted, cracked, chipped, peeled, swollen, or eroded components.

3.8.2 Low temperature. The test set shall comply with the requirements specified in 3.5, 3.6, 3.7 and the applicable drawings while operating at -65° ± 5°F. when tested as specified in 4.4.4.2. After the completion of the test the test set shall show no evidence of deterioration or damage, such as obliterated markings, embrittled, distorted, cracked, chipped, peeled, swollen, or eroded components.

3.8.3 Vibration. The test set shall show no evidence of chipped, broken, or loose components and shall meet the requirements of 3.5, 3.6, 3.7, while subjected to vibrations through the frequencies, amplitude, and time when tested as specified in 4.4.4.3.

3.9 Non-operating environment testing.

3.9.1 High temperature. The test set shall show no evidence of deterioration or damage, such as obliterated markings, embrittled, distorted, cracked, chipped, peeled, swollen, or eroded components and shall meet the requirements of 3.5, 3.6, and 3.7 after subjection to 165° ± 5°F when tested as specified in 4.4.5.1.

3.9.2 Low temperature. The test set shall show no evidence of deterioration or damage, such as obliterated markings, embrittled, distorted, cracked, chipped, peeled, swollen, or eroded components and shall meet the requirements of 3.5, 3.6, and 3.7 after subjection to -65° ± 5°F when tested as specified in 4.4.5.2.

3.9.3 Altitude. The test set shall comply with the requirements as specified in 3.5, 3.6, and 3.7 after being tested as specified in 4.4.5.3.

3.9.4 Drop. The test set and auxiliaries shall show no evidence of damage and the test set shall meet the requirements of 3.5, 3.6, and 3.7 after being tested as specified in 4.4.5.4.

3.9.5 Humidity. The test set and auxiliaries shall show no evidence of deterioration or damage, such as obliterated markings, embrittled, distorted, cracked, chipped, peeled, swollen, or eroded components. The relative humidity reading of the indicator on the auxiliaries shall be 40 percent maximum: the test set shall meet the requirements of 3.5, 3.6, and 3.7 after being tested as specified in 4.4.5.5.

3.10 Leakage. The auxiliaries shall show no evidence of leakage when tested as specified in 4.4.6.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

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3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwgs 8803976 and 8803977) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2. Inspection and testing to be performed. The first article inspecion sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (see 3.5)		X

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3. Dielectric strength (see 3.6)		X
4. Insulation resistance (see 3.7)		X
5. High temperature (operating) (see 3.8.1)		X
6. Low temperature (operating) (see 3.8.2)		X
7. Vibration (see 3.8.3)	X	
8. High temperature (non-operating) (see 3.9.1)		X
9. Low temperature (non-operating) (see 3.9.2)		X
10. Altitude (see 3.9.3)		X
11. Drop (see 3.9.4)	X	
12. Humidity (see 3.9.5)	X	
13. Leakage (see 3.10)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Unit assembly (see dwg 8785831).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper location of electrical component	Visual
	102. Improper location of name plate	Visual
	103. Damaged wiring, such as abrasion, break, etc	Visual
	104. Damaged electrical components	Visual/Manual
	105. Hardware and fastener not installed properly and securely	Visual/Manual
	106. Loose wiring connection	Visual/Manual
Minor:		
	201. Paint damaged	Visual
	202. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 Assembly with cover removed prior to installing liner (see dwg 8804125).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Shock mount not fastened securely	Visual

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102. Shock mount not oriented properly.....Visual
103. Humidity indicator not fastened securely.....Visual

Minor:
201. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.3 Assembly with cover removed prior to installing unit assembly (see dwg 8804125).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Liner fastened insecurely.....	Visual/Manual
	102. Dehydrator fastened insecurely.....	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.11).....	Visual

4.3.2.1.4 Assembly prior to securing cover (see dwg 8803976).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Unit assembly fastened improperly.....	Visual
	102. Damage to desiccant bag, such as tear, hole, etc.....	Visual
	103. Desiccant bag not in place.....	Visual
	104. Sealing gasket containing foreign matter.....	Visual
	105. Improper engagement of camloc fastener.....	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.11).....	Visual

4.3.2.1.5 Assembly, test set (package 1 of 2) (see dwg 8803976).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101.	Damaged case	Visual
102.	Cover fits improperly	Visual
103.	Seal not intact	Visual
104.	Marking incorrect, illegible, missing	Visual
105.	Packing list not in envelope	Visual
106.	Catches insecure and disengaged	Visual/Manual
107.	Handle not secure	Visual/Manual
108.	*Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
109.	**Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual

Minor:

201.	Evidence of poor workmanship (see 3.11)	Visual
202.	Envelope fastened insecurely	Visual

4.3.2.1.6 Auxiliaries with cover removed (see dwg 8803977).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Damaged container	Visual
102.	Improper packing and packaging	Visual
103.	Marking, incorrect, missing, illegible	Visual
104.	Packing list missing	Visual
105.	Missing component	Visual
106.	Humidity indicator missing or loose	Visual/Manual
107.	*Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
108.	**Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual

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Minor:
201. Evidence of poor workmanship (see 3.11).....Visual
202. Envelope fastened insecurely.....Visual

4.3.2.1.7 Auxiliaries with cover attached (see dwg 8803977).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Container improperly sealed.....	Visual
	102. Cover loose.....	Visual/Manual
	103. Objective evidence relative to material missing or improper (see 3.1).....	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11).....	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, a Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to adjacent sockets, a Gage (see 4.3.4) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

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4.3.2.1.8 Preparation for delivery (see dwgs 8803976 and 8803977).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packaging, packing and preservation in accordance with dwgs 8803976 and 8803977	Visual
	102. Marking in accordance with dwgs 8803976 and 8803977	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity	Major
b. Dielectric strength	Major
c. Insulation resistance	Major
d. Leakage (auxiliaries)	Major
e. High temperature	Major
f. Low temperature	Major
g. Vibration	Major

Of these tests, g is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity, dielectric strength, insulation resistance, and leakage. Each test set or auxiliaries as applicable shall be tested in accordance with 4.4.1 through 4.4.3 and 4.4.6. If any item fails to comply with 3.5 through 3.7 and 3.10, the item shall be rejected and removed from the lot.

4.3.3.2 High temperature, low temperature, and vibration. One sample test set or auxiliaries as applicable shall be selected at random by the Government inspector from each 25 consecutively produced units. The sample shall be tested in accordance with 4.4.4.1, through 4.4.4.3. If the sample fails to comply with 3.8.1 through 3.8.3, the lot shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and test. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8815532). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.4.1 Continuity. The continuity of the test set set shall be checked by utilizing Electrical Test Panel, drawing 8815011 (see Figure 1), or its equivalent.

4.4.2 Dielectric strength. Each ungrounded contact shall be subjected to the required ac potential between the connector and electrically isolated contacts. The electrical test panel shall be in accordance with Figure 1.

4.4.2.1 Preliminary set-up. All switches on the test set shall be in the "Off" position with the exception of S2, which shall be in the "FB" position, S9 which shall be in the "On" position and S3 which shall be in the neutral position. Jump all diodes. Jump J7H and J7G.

4.4.2.2 Test procedure for Figure 1. Place switches S14 through S17 to the "1" position, and place switches S18 through S21 to the "2" position. Attach clip #1 to E_1 ground. Connect approved dielectric strength testing equipment (see 6.3) to E_1 and E_2 terminals of the electrical test panel (see Figure 1). Rotate switches S10 through S13 through all positions.

4.4.3 Insulation resistance. Each ungrounded contact shall be subjected to the required dc potential between the connector and electrically-isolated contacts. The electrical test panel shall be in accordance with Figure 1. Test procedure shall include 4.4.2.1 and 4.4.2.2 with the exception that an approved megger shall be used in place of the dielectric strength tester.

4.4.4 Operating environment tests (see 3.8).

4.4.4.1 High temperature. The test set with cover removed and in normal operating position shall be placed in a chamber and stabilized (see 6.4) at 165° ± 5°F. with a maximum relative humidity of 15 percent. While at this temperature the test set shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3.

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4.4.4.2 Low temperature. The test set with cover removed and in normal operating position shall be placed in a chamber and stabilized at -65° + 5°F with a relative humidity that will produce frozen condensate. While at this temperature the test set shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3.

4.4.4.3 Vibration. The test set while operating and auxiliaries with cover attached shall be secured to a vibration table and vibrated for 45 minutes in each of three mutually perpendicular directions through a frequency range of 10 to 55 Hz at a double amplitude of 0.063 inch. The frequency shall be varied uniformly from 10 to 55 to 10 Hz making approximately one cycle per minute. After completion of this test the test set shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3, the cover shall be removed from the auxiliaries, and the container shall meet the packaging requirements of dwg 8803977. Following this test the cables in the container shall meet the continuity, dielectric strength, and insulation resistance requirements of MIL-C- (PA).

4.4.5 Non-operating environment test (see 3.9).

4.4.5.1 High temperature. The test set with cover attached shall be placed in a test chamber and the temperature stabilized at 165° + 5°F for ten hours. The test set then shall be reduced to ambient room temperature and tested as specified in 4.4.1, 4.4.2 and 4.4.3.

4.4.5.2 Low temperature. The test set with cover attached shall be placed in a test chamber and the temperature stabilized at -65° + 5°F for ten hours. The test set then shall be raised to ambient room temperature and tested as specified in 4.4.1, 4.4.2 and 4.4.3.

4.4.5.3 Altitude. The test set with cover attached shall be placed in a test chamber and stabilized at a pressure of 2.3 inches of mercury absolute for two hours and then returned to room barometric pressure. It shall then be tested as specified in 4.4.1, 4.4.2 and 4.4.3.

4.4.5.4 Drop. The test set and auxiliaries with cover attached shall be dropped from a height of 30 inches onto a steel, concrete or stone surface of sufficient mass to absorb the shock without deflection in such a manner that the face of the equipment absorbs the full force of the fall. (The height of 30 inches refers to the distance from the steel, concrete, or stone surface to the nearest face of the containers when suspended prior to the fall). The fall shall be a free fall in that no ropes or other suspending media are attached to the equipment during the fall. One drop shall be made for each face or side of the test set transit case and each face or side of the auxiliaries container. After completion of this test the test set shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3; the cover shall be removed from the auxiliaries, and the container shall meet

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the packaging requirements of dwg 8803977. Following this the cables in the container shall meet the continuity, dielectric strength, and insulation resistance requirements of MIL-C- (PA).

4.4.5.5 Humidity. The test set with the cover attached and auxiliaries shall be placed in a suitable humidity chamber and subjected to five humidity cycles. While the test set and auxiliaries (package 2) are in the chamber, the humidity shall be maintained at 90 to 98 percent relative humidity throughout the following cycle. The temperature shall be raised to 149°F + 5°F within four hours and maintained at this point for eight hours. The temperature shall then be reduced to 85°F + 5°F within four hours and maintained at this point for 2I hours. The temperature shall be reduced to 68°F + 5°F within one hour and maintained at this point for four hours. The temperature shall then be raised to 86°F + 5°F within one hour and maintained at this point for five hours. At the end of this test the test set shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3. After completion of this test the cover shall be removed from the auxiliaries, and the container shall meet the packaging requirements of dwg 8803977. Following this test the cables in the container shall meet the continuity, dielectric strength, and insulation resistance requirements of MIL-C-45889(PA).

4.4.6 Leakage. The equipment shall be set-up with the test chamber connected to an air supply and a measuring device (manometer) as shown in Figure 2. The auxiliaries shall be placed in the test chamber and the chamber pressurized until a reading of 4.62 feet of water is reached in the manometer tube. The air supply is then turned off and the water level in the manometer tube is observed for thirty seconds for indications of a pressure drop. Any pressure drop occurring during the thirty second test period shall not exceed the calibrated pressure drop for the system.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set and auxiliaries shall be packaged in accordance with dwgs 8803976 and 8803977 using the specified moisture barrier.

5.1.2 Level C. The test set and auxiliaries shall be packaged in accordance with dwgs 8803976 and 8803977 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set and auxiliaries shall be packed in accordance with dwgs 8803976 and 8803977.

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5.3 Marking. Marking shall be as specified in accordance with dwgs 8803976 and 8803977.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 5.4 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

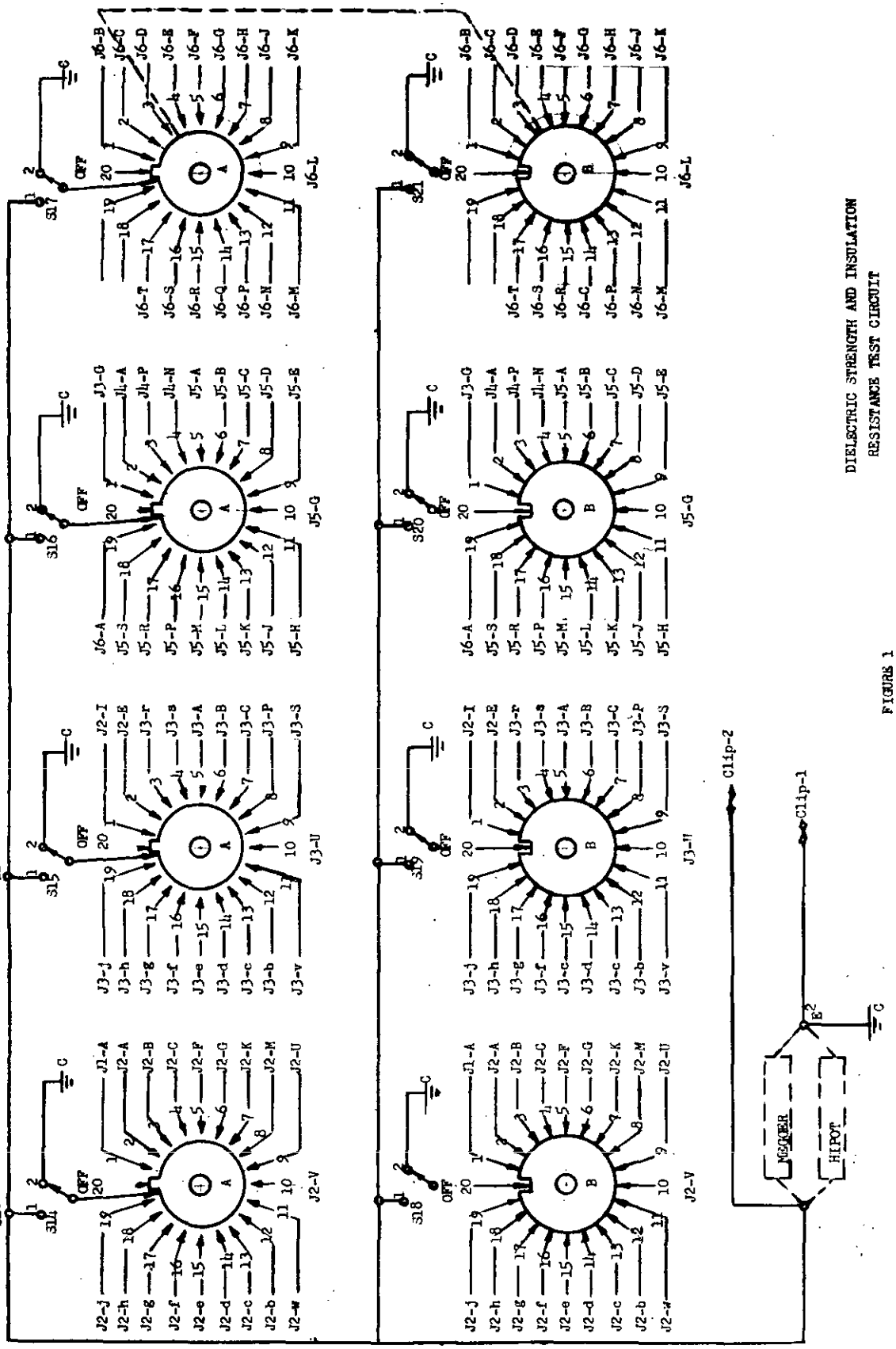
6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1048 dated 13 June 1958.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A083

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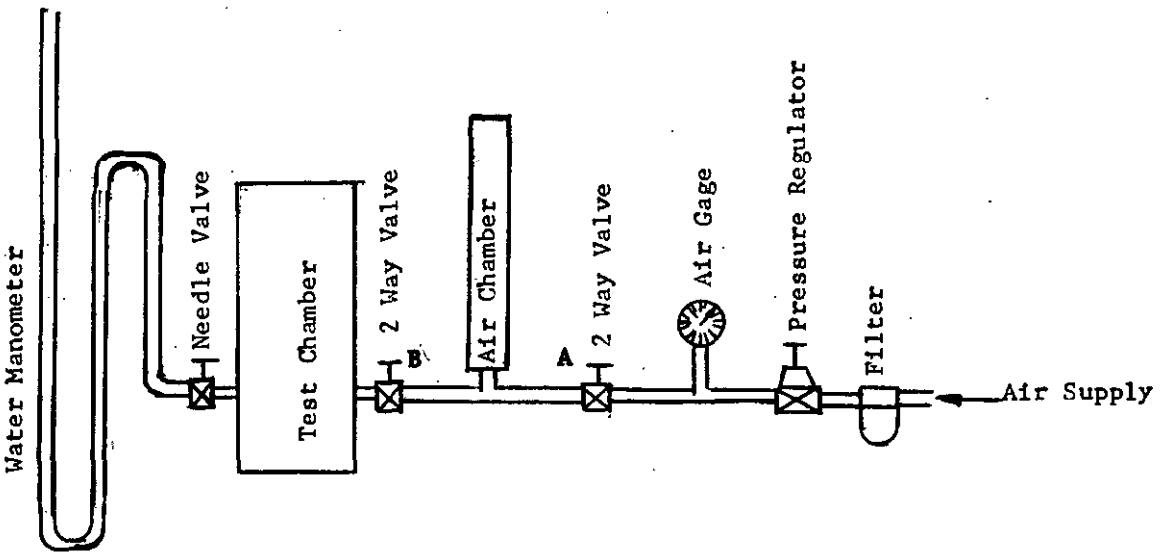


DIELECTRIC STRENGTH AND INSULATION
RESISTANCE TEST CIRCUIT

FIGURE 1

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Diameter and length of test chamber should be held to a maximum of 1" larger than the container to be tested.



1. Place a block of wood, conforming to the approximate size and shape of the item, in the test chamber and perform a system leakage calibration test per steps 2 through 8.

2. Adjust pressure regulator to a certain pressure depending on the size of air chamber and test chamber (2 psi pressure is required in test chamber).

3. Open valve A and allow air chamber to fill.

4. Close Valve A.

5. Open Valve B and allow air from air chamber to equalize. Pressure should be 2 psi.

6. Manometer should read 4.62 feet of water.

7. Manometer should maintain a constant level for 30 seconds. Drop in pressure will indicate a leak. Record leakage value if any. If the system leakage is above .25 p.s.i. the system leakage will have to be lowered prior to performing further tests.

8. Close Valve B and remove item.

9. Place item to be tested in the test chamber and repeat steps 2 through 8. If any leakage is recorded above that recorded in step 7 the item shall be rejected.

Figure 2

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ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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